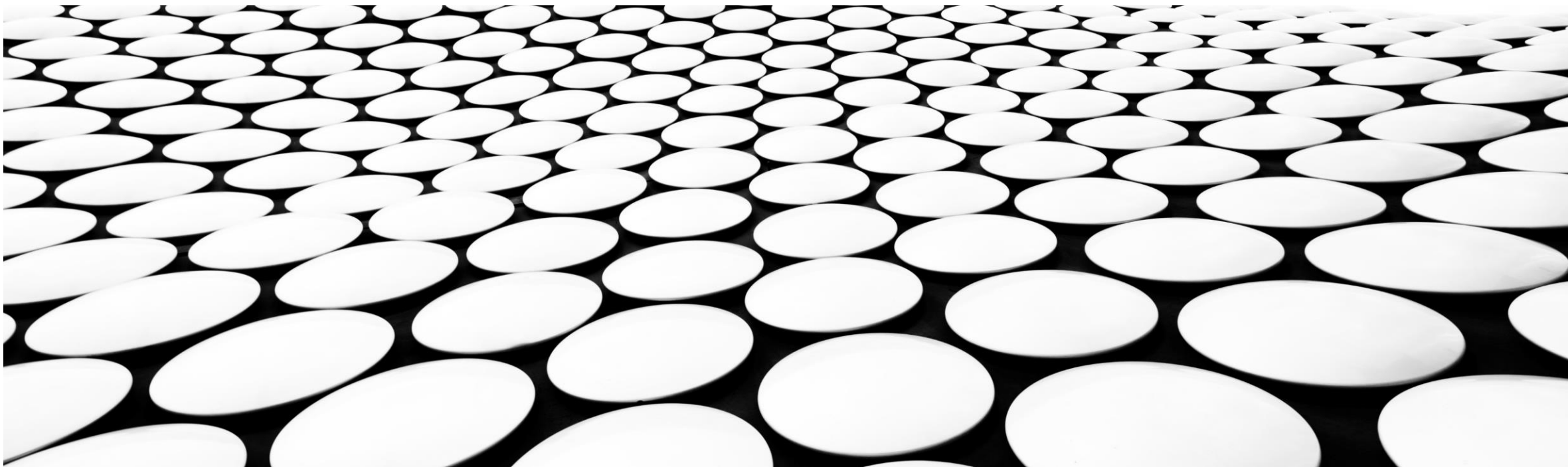

AI CODING-01 : 簡易工單系統與閉環管理

USE DEEPSEEK V4 FLASH ON OPENCODE

[HTTPS://GITHUB.COM/ALENGBK/PEPPERMINT-PYTHON](https://github.com/ALENGBK/PEPPERMINT-PYTHON)



DASHBOARD

Peppermint-fork

v1.0.0

Dashboard

Tickets

ADMINISTRATION

Users

Clients

Audit Log

API Keys

admin

EN | 中

Sign out

Dashboard

New Ticket

67

Open Issues

1

Completed Issues

68

Total Tickets

Recent Tickets

Export CSV

TICKET 主畫面

Peppermint-fork
v1.0.0

Dashboard

Tickets

ADMINISTRATION

Users

Clients

Audit Log

API Keys

admin
EN | 中 Sign out

Tickets

New Ticket

Export CSV

All Status

All Priority

Search tickets...

#	TITLE	PRIORITY	STATUS	TYPE	ASSIGNEE
#68	EC2 資源異動警告 ×130 API	Medium	Closed	Incident	—
#67	EC2 資源異動警告 API	Medium	Closed	Incident	—
#66	allen post api API	Low	Open	Incident	D
#65	TLS 版本不相容? Human	urgent	Open	access	—
#64	印表機沒反應! Human	urgent	Open	Feature	—
#63	防火牆規則衝突? Human	Low	Open	Bug	—
#62	Redis 快取擊穿? Human	urgent	Open	service	—
#61	TLS 版本不相容? Human	High	Open	Incident	—
#60	Webhook 未送達! Human	High	Open	access	—
#59	容器重啟循環 Human	Low	Open	maintenance	—
#58	容器重啟循環 Human	Low	Open	Bug	—
#57	磁碟 I/O 瓶頸! Human	High	Open	Incident	—
#56	OOM Killer 觸發 Human	High	Open	access	—
#55	憑證即將到期? Human	urgent	Open	service	—
#54	VPN 連線中斷? Human	Low	Open	Support	—

TICKET 內容

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Dashboard

Tickets

ADMINISTRATION

Users

Clients

Audit Log

API Keys

admin

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Sign out

Ticket Detail

Back to Tickets

Delete

#68

EC2 資源異動警告

Medium

Closed

Incident

API

[RED] 資源數量確認 : EC2 有变动
新增> "i-0d98f732f779fc9c5"

Created: 2026/9/4 上午09:02 by api

Activity

admin 2026/9/7 上午01:46

config 檔案配置錯誤

Add a comment...

Comment

Re-open

ASSIGNEE

Unassigned

PRIORITY

Medium

STATUS

Closed

ALERT COUNT

× 130

Last seen: 2026/9/7 上午01:32

#1	2026/9/4 下午05:00
#2	2026/9/4 下午05:32
#3	2026/9/4 下午06:02
#4	2026/9/4 下午06:32
#5	2026/9/4 下午07:02
#6	2026/9/4 下午07:32
#7	2026/9/4 下午08:02
#8	2026/9/4 下午08:32
#9	2026/9/4 下午09:02

使用者管理

Peppermint-fork
v1.0.0

Dashboard

Tickets

ADMINISTRATION

Users

Clients

Audit Log

API Keys

admin

EN | 中 Sign out

Users

New User

Search users...

NAME	EMAIL	ROLE	CREATED	ACTIONS
D david	david@aa.local	Agent	2026/7/29 上午07:07	Edit Delete
A admin	admin@peppermint.com	Admin	2026/7/27 上午03:53	Edit Delete

API KEY 管理

Peppermint-fork
v1.0.0

Dashboard

Tickets

ADMINISTRATION

Users

Clients

Audit Log

API Keys

admin

EN | 中

Sign out

Service Accounts

New API Key

NAME	DESCRIPTION	KEY PREFIX	STATUS	LAST USED	CREATED BY	ACTIONS
api		pep_c7504d551a27f7c7...	Active	2026/9/4 上午09:02	admin@peppermint.com	<button>Regenerate</button> <button>Revoke</button>

AUDIT LOG

Peppermint-fork

v1.0.0

Dashboard

Tickets

ADMINISTRATION

Users

Clients

Audit Log

API Keys

admin

EN | 中

Sign out

Audit Log

All Actions

All Sources

Search email...

Total: 117

TIME	EMAIL	ACTION	DETAIL	SOURCE	IP ADDRESS
2026/9/4 上午09:27	admin@peppermint.com	Login	Login successful	Human	10.14.21.248
2026/9/4 上午09:02	api	Ticket Created	Ticket #68: EC2 資源異動警告	API	
2026/9/4 上午08:57	admin@peppermint.com	Ticket Closed	Ticket #67: EC2 資源異動警告	Human	
2026/9/4 上午08:57	admin@peppermint.com	Comment	Comment on ticket 614c9572-d651-4f79-a8e5-d23fa9a67dbc	Human	
2026/9/4 上午08:55	api	Ticket Created	Ticket #67: EC2 資源異動警告	API	
2026/9/4 上午08:48	admin@peppermint.com	Deleted	Deleted ticket #70: EC2 資源異動警告	Human	
2026/9/4 上午08:47	admin@peppermint.com	Deleted	Deleted ticket #69: \$title	Human	
2026/9/4 上午08:47	admin@peppermint.com	Deleted	Deleted ticket #68: EC2	Human	
2026/9/4 上午08:47	admin@peppermint.com	Deleted	Deleted ticket #67: EC2	Human	
2026/9/4 上午08:47	api	Ticket Created	Ticket #70: EC2 資源異動警告	API	
2026/9/4 上午08:16	api	Ticket Created	Ticket #69: \$title	API	
2026/9/4 上午08:08	api	Ticket Created	Ticket #68: EC2	API	
2026/9/4 上午08:07	api	Ticket Created	Ticket #67: EC2	API	
2026/9/4 上午07:51	admin@peppermint.com	Deleted	Deleted ticket #67: EC2	Human	
2026/9/4 上午07:51	admin@peppermint.com	Deleted	Deleted ticket #68: EC2	Human	

CSV 匯出

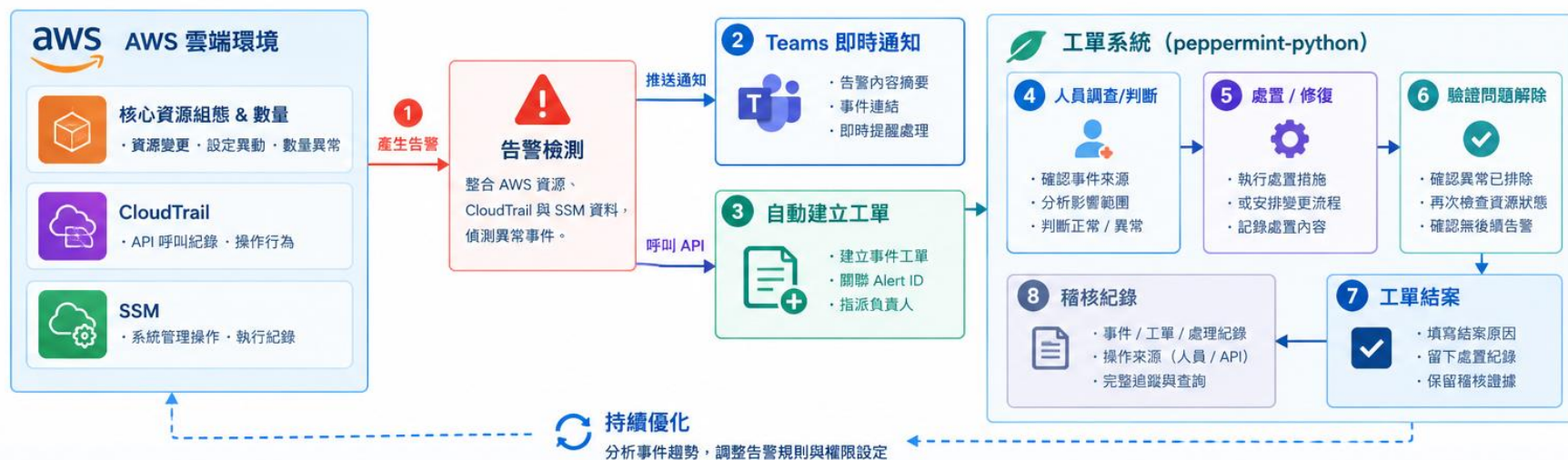
```
tickets.csv X
C: > Users > allenchiu > Downloads > tickets.csv
1  Number,Title,Status,Priority,Type,Assignee Name,Assignee Email,Client,Created At,Updated At
2  68,EC2 資源異動警告,needs_support,medium,incident,,,,,2026-09-04T09:02:22.010097,2026-09-04T09:02:22.010097
3  67,EC2 資源異動警告,done,medium,incident,,,,,2026-09-04T08:55:52.602682,2026-09-04T08:57:46.122191
4  66,allen post api,needs_support,low,incident,david,david@aa.local,,2026-07-29T07:06:56.520240,2026-07-29T07:08:09.237094
5  65,TLS 版本不相容?,needs_support,urgent,access,,,,,2026-07-27T05:34:19.932715,2026-07-27T05:34:19.932715
6  64,印表機沒反應!,needs_support,urgent,feature,,,,,2026-07-27T05:34:19.923793,2026-07-27T05:34:19.923793
7  63,防火牆規則衝突?,needs_support,low,bug,,,,,2026-07-27T05:34:19.915701,2026-07-27T05:34:19.915701
8  62,Redis 快取擊穿?,needs_support,urgent,service,,,,,2026-07-27T05:34:19.910113,2026-07-27T05:34:19.910113
9  61,TLS 版本不相容?,needs_support,high,incident,,,,,2026-07-27T05:34:19.904128,2026-07-27T05:34:19.904128
10 60,Webhook 未送達!,needs_support,high,access,,,,,2026-07-27T05:34:19.898328,2026-07-27T05:34:19.898328
11 59,容器重啟循環,needs_support,low,maintenance,,,,,2026-07-27T05:34:19.892212,2026-07-27T05:34:19.892212
12 58,容器重啟循環,needs_support,low,bug,,,,,2026-07-27T05:34:19.885502,2026-07-27T05:34:19.885502
13 57,磁碟 I/O 瓶頸!,needs_support,high,incident,,,,,2026-07-27T05:34:19.877859,2026-07-27T05:34:19.877859
14 56,OOM Killer 觸發,needs_support,high,access,,,,,2026-07-27T05:34:19.869891,2026-07-27T05:34:19.869891
15 55,憑證即將到期?,needs_support,urgent,service,,,,,2026-07-27T05:34:19.863572,2026-07-27T05:34:19.863572
16 54,VPN 連線中斷?,needs_support,low,support,,,,,2026-07-27T05:34:19.857432,2026-07-27T05:34:19.857432
17 53,路由表收斂異常,needs_support,medium,incident,,,,,2026-07-27T05:34:19.851283,2026-07-27T05:34:19.851283
18 52,S3 上傳失敗!,needs_support,medium,service,,,,,2026-07-27T05:34:19.845098,2026-07-27T05:34:19.845098
19 51,排程任務未觸發?,needs_support,high,access,,,,,2026-07-27T05:34:19.837383,2026-07-27T05:34:19.837383
20 50,DNS 解析失敗?,needs_support,medium,service,,,,,2026-07-27T05:34:19.830475,2026-07-27T05:34:19.830475
21 49,憑證即將到期!,needs_support,medium,access,,,,,2026-07-27T05:34:19.817469,2026-07-27T05:34:19.817469
22 48,印表機沒反應?,needs_support,urgent,maintenance,,,,,2026-07-27T05:34:19.811411,2026-07-27T05:34:19.811411
23 47,LDAP 同步失敗!,needs_support,urgent,feature,,,,,2026-07-27T05:34:19.805051,2026-07-27T05:34:19.805051
24 46,資料表索引損毀,needs_support,medium,maintenance,,,,,2026-07-27T05:34:19.798833,2026-07-27T05:34:19.798833
25 45,VPN 連線中斷?,needs_support,urgent,bug,,,,,2026-07-27T05:34:19.791154,2026-07-27T05:34:19.791154
26 44,TLS 版本不相容,needs_support,low,support,,,,,2026-07-27T05:34:19.782432,2026-07-27T05:34:19.782432
27 43,SSH 連線被拒,needs_support,medium,incident,,,,,2026-07-27T05:34:19.776232,2026-07-27T05:34:19.776232
28 42,BGP session 中斷,needs_support,urgent,maintenance,,,,,2026-07-27T05:34:19.770617,2026-07-27T05:34:19.770617
29 41,備份排程未執行,needs_support,low,service,,,,,2026-07-27T05:34:19.765185,2026-07-27T05:34:19.765185
30 40,Syslog 未轉發,needs_support,urgent,bug,,,,,2026-07-27T05:34:19.759425,2026-07-27T05:34:19.759425
31 39,Webhook 未送達?,needs_support,low,maintenance,,,,,2026-07-27T05:34:19.752535,2026-07-27T05:34:19.752535
32 38,記憶體使用率過高!,needs_support,high,feature,,,,,2026-07-27T05:34:19.744012,2026-07-27T05:34:19.744012
33 37,檔案伺服器空間不足?,needs_support,urgent,service,,,,,2026-07-27T05:34:19.735504,2026-07-27T05:34:19.735504
```


SUMMARY

AWS 雲端資安告警閉環管理架構

從異常偵測到事件結案，串接 Teams 與工單系統，建立可追蹤、可稽核的資安告警閉環流程

- ✓ 即時通知
- ✓ 自動建單
- ✓ 人員確認
- ✓ 提高處理效率
- ✓ 完整稽核紀錄
- ✓ 持續改善



流程說明

- 1 AWS 資源異動產生告警**
(組態、數量、操作行為)
- 2 Teams 即時通知**
(告警內容與連結)
- 3 自動建立工單**
(關聯 Alert ID，指派負責人)
- 4 人員調查 / 判斷**
(確認來源與影響，判斷正常 / 異常)
- 5 處置 / 修復**
(執行措施或安排變更)
- 6 驗證問題解除**
(確認無異常、無後續告警)
- 7 工單結案**
(填寫原因、記錄處置結果)
- 8 稽核紀錄**
(完整追蹤、可查詢)

工單系統關鍵功能

- ✓ 支援 API 建單與狀態管理
 - ✓ 指派負責人與團隊
 - ✓ 留言 / 處理紀錄
 - ✓ 時間追蹤
 - ✓ 稽核日誌 (人員 / API 來源)
 - ✓ 支援 Alert ID 關聯、結案原因與證據紀錄
- peppermint-python
(AI Coding 建置的簡易工單系統)

目前成果與後續規劃

- ✓ **現況成果**
 - AWS 告警 + Teams 即時通知
 - 自動建立工單並串接處理流程
 - 完成 Alert → Ticket → Investigation → Closure 的閉環
- » **後續規劃**
 - 新增 Alert ID、Closure Reason、Evidence 欄位
 - 導入 SLA 與逾時升級機制
 - 依事件類型評估自動化處置 (SOAR)

目前 AWS 雲端告警除了透過 Teams 即時通知外，

我另外建立了告警到工單的串接機制。

當偵測到 AWS 資源組態、數量或操作異動時，系統會自動建立工單，保留事件來源、處理人員與處置紀錄，再由管理人員確認後結案。

目前主要採 Human-in-the-loop 的方式，先完成 Alert → Ticket → Investigation → Closure 的閉環，後續再視事件類型導入自動化處置。